Deep Learning-Based Monocular Visual Odometry for Autonomous Spatial Mapping

Synopsis report on
Deep Learning Project
[ICT-4442]

Submitted By

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SYNOPSIS

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Title of the Project: Deep Learning Based Monocular Visual Odometry for Autonomous Spatial Mapping

Description:

Monocular Visual Odometry (MVO) is the estimation of 6-DoF ego-motion of a single camera using a sequence of images by estimating the relative rotation and translation between consecutive frames. This work replaces the existing fragile pipelines of geometric visual odometry based on hand-engineered methods with novel deep-learning based architectures which map pixel intensity changes to spatial movements – important for navigation in GPS denied areas (indoor warehouses, urban canyons, extraterrestrial planets) where autonomous vehicles and robots depend only on their vision system. Four novel architectural solutions will be implemented: (1) classical ORB/Lucas-Kanade tracker + MLP baseline; (2) CNN+GRU/LSTM recurrent architecture for temporal consistency; (3) self-supervised PoseNet+DepthNet autoencoder model trained through photometric reprojection loss; and (4) CNN + CBAM/SE Attention model. All four models will be trained and tested on Virtual KITTI 2 using same train/val/test split and metrics (ATE, RPE, depth RMSE for Model 3) with monocular scale ambiguity dealt with through ground truth scaling (for Models 1, 2, 4) or automatic scaling from depth (Model 3).

Dataset:

Virtual KITTI 2 (vKITTI 2) – A synthetic photo-realistic driving dataset made using Unity 3D, substitute for real world KITTI. Offers RGB frames, precise 6 DoF poses, optical flow, and dense depth map generation of 100% (compared to about 5% by physical LiDAR sensors). Since this is synthetic data, performance will be at an ideal upper bound level; domain difference between real and synthetic will be highlighted as a drawback. Data Preprocessing: resized to 128 x 416, ImageNet Normalization, temporal stacking (2-frames spatial & 5-10 frame recurrent).

Proposed Models for Comparison:

- Model 1 - Classical ORB/Lucas-Kanade feature tracking + MLP regressor (classical/MLP baseline).
- Model 2 - CNN (ResNet) + GRU/LSTM recurrent pose regression (RNN / temporal sequence modeling).
- Model 3 - Self-supervised PoseNet + DepthNet autoencoder trained via photometric reprojection loss (unsupervised learning).
- Model 4 - CNN + CBAM/SE-Net convolutional attention for spatial/channel feature weighting (attention mechanism).

Model-to-Member Assignment:

- Member 1 (Vaibhav Krishna Bansal) - Model 4: CNN + Convolutional Attention (CBAM/SE-Net).
- Member 2 (Pranav Joshi) - Model 1: Classical Feature Extraction + MLP Baseline.
- Member 3 (Abhishek Laxman Jadhav) - Model 2: CNN + GRU/LSTM Recurrent Architecture.
- Member 4 (Suryansh Bajaj) - Model 3: Self-Supervised PoseNet/DepthNet Autoencoder.

Existing Methods:

- CoProU-VO (Xie et al., 2025) - cross-frame uncertainty masking for unsupervised VO in dynamic scenes.
- SfM-Learner (Zhou et al., 2017) / Godard et al. (2019) - early work on self-supervised depth+pose learning.
- DeepVO (Wang et al., 2017) - End-to-end CNN+LSTM recurrent VO.
- LKN-VO (Peng et al., 2019) - learned Kalman-style filtering.

Objectives:

- Achieve accurate 6-DoF ego-motion estimation from monocular image sequences without GPS or auxiliary sensors.
- Quantify accuracy, robustness, and computational trade-offs across classical, recurrent, self-supervised, and attention-based architectures for visual odometry.
- Establish a common, fair evaluation protocol - identical dataset splits and metrics - enabling direct, reproducible comparison across all four models.

Importance / Motivation of the Proposed Project:

Navigation without GPS is essential for UAVs and robots that operate inside buildings, in urban canyons, or in extraplanetary missions. Geometric visual odometry fails if there is motion blur, illumination changes, and dynamic elements in the images. This survey of deep learning approaches provides insight into which architectural components increase robustness, which directly informs the design of the team's own UAV perception pipeline.

Preliminary Reference List:

[1] V. Cabon et al., "Virtual KITTI 2," arXiv:2001.10773, 2020.

[2] J. Xie et al., "CoProU-VO: Combining Projected Uncertainty for Unsupervised MVO," GCPR, 2025.

[3] S. Wang et al., "DeepVO: End-to-End Visual Odometry with Deep RCNNs," IEEE ICRA, 2017.

[4] T. Zhou et al., "Unsupervised Learning of Depth and Ego-Motion from Video," IEEE CVPR, 2017.

[5] C. Godard et al., "Digging Into Self-Supervised Monocular Depth Estimation," IEEE ICCV, 2019.

[6] H. Peng et al., "Learning Kalman Network: Deep MVO for On-Road Driving," Rob. Auton. Syst., 2019.

[7] "MINI-DROID-SLAM: Improving Monocular SLAM Using MINI-GRU RNN," PMC Open Access, 2024.

[8] "An Unsupervised MVO Based on Multi-Scale Modeling," Sensors (MDPI), 2022.

[9] "Reinforcement Learning Meets Visual Odometry," ECCV, 2024.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	Finalize problem statement, dataset, architecture assignments; submit Synopsis	25 Aug 2026
2	Set up GitHub repo; complete vKITTI 2 preprocessing pipeline	10 Sep 2026
3	Implement ≥2 models with preliminary results; complete 8-10 paper lit. review; submit Interim report	25 Sep 2026
4	Complete remaining models; hyperparameter tuning; unified comparison table	20 Oct 2026
5	Presentations and Q&A; submit Final report + code repository	26-31 Oct 2026